OpenClaw AI Rock-Paper-Scissors Game Tutorial

> Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests] in the `Pre-use Preparation` directory. If you are only using TUI / WhatsApp, you can skip `05` for now. This article assumes the basic environment is already prepared and only covers the AI Rock-Paper-Scissors project itself.

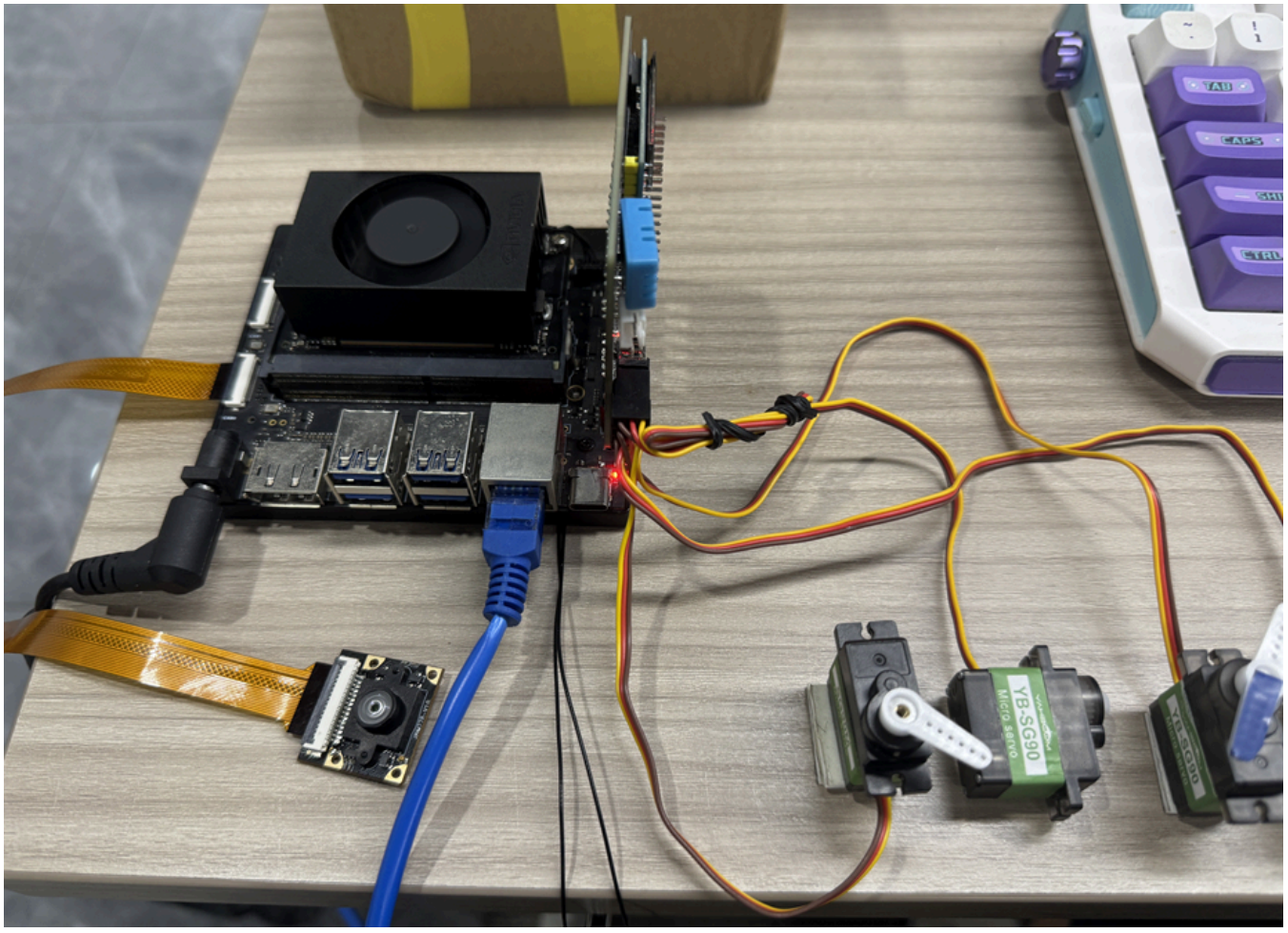
1. Tutorial Objective

This article demonstrates OpenClaw's interactive gaming capability through "gesture recognition + AI referee + screen and light feedback". After completing this tutorial, readers can:

- Learn how to use the camera to recognize player gestures
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice code
- Complete four project scenarios: gesture recognition, AI move, result determination, score feedback

2. Hardware Preparation

- OpenClaw main control board
- CSI camera module
- OLED display
- RGB light strip / RGB LED module
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Camera capture command: `/home/jetson/hardware_hal_demo/seewhat`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The key capabilities of this project mainly consist of three parts:

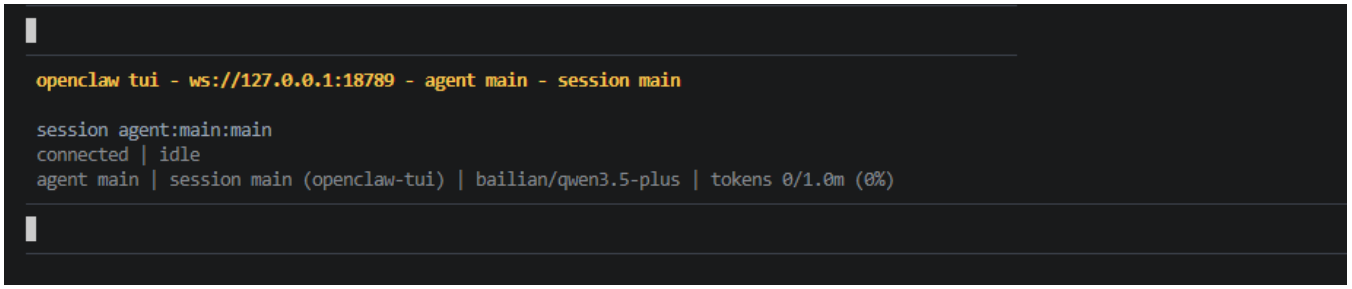
- The camera is responsible for capturing the player's gesture
- OpenClaw is responsible for deciding the AI's move and displaying the result on OLED / RGB

4. Testing Basic Project Capabilities

Before starting this project, please complete an `openclaw tui` basic dialogue self-check first; if you haven't done so, see [06-Three Dialogue Configuration Tests] first. After passing, use TUI to verify the basic capabilities in the project.

Enter in terminal:

```
openclaw tui
```



Then you can start with a few basic tests:

- Take a photo of my gesture
- Check if this is a scissors gesture
- Display "Ready" on the OLED
- Light the RGB blue

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
WhatsApp	Remote demonstrations, event interactions	Easy to share with multiple people	Demonstrate "visual interactive game"
TUI	Local debugging, rule verification	Easiest to observe the recognition and determination process	First run through game logic
Voice code	Game interaction, classroom demonstrations	More like a real AI game partner	Use as complete demonstration solution

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.

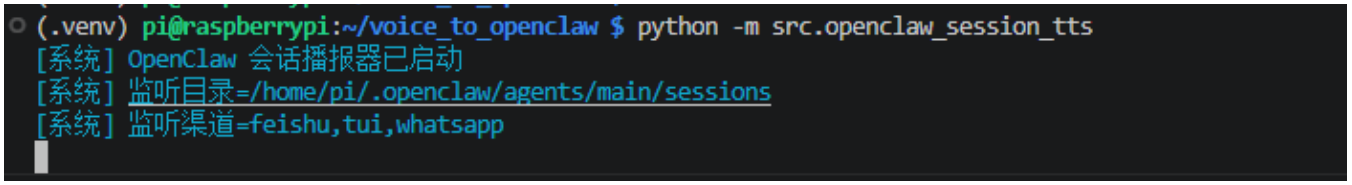
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [06-Three Dialogue Configuration Tests]. After creation, you can remotely trigger the Rock-Paper-Scissors game in WhatsApp, for example:

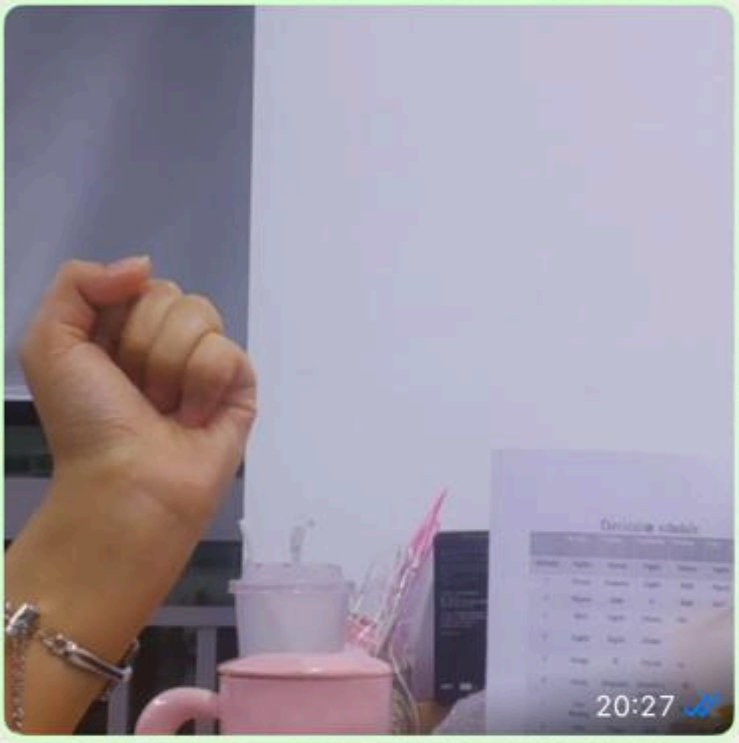
- Let's play Rock-Paper-Scissors, take a photo to recognize my gesture, you can randomly make a gesture, then judge whether you won or I won, OLED displays who won this round

20:27

< 2 +86 153 3885 7526 (你)
给自己发消息

Please play rock paper scissors. Take a photo to recognize my gesture. You can randomly make a gesture.Then judge whether you win or I win OLED displays who this round.

20:27



I can see the user's hand gesture in the image - it's a **fist**, which represents **ROCK** in rock paper scissors.

Now I need to:

1. Make a random gesture choice (rock, paper, or scissors)
2. Determine the winner
3. Display the result on OLED



You can add your own WhatsApp dialogue screenshots here.

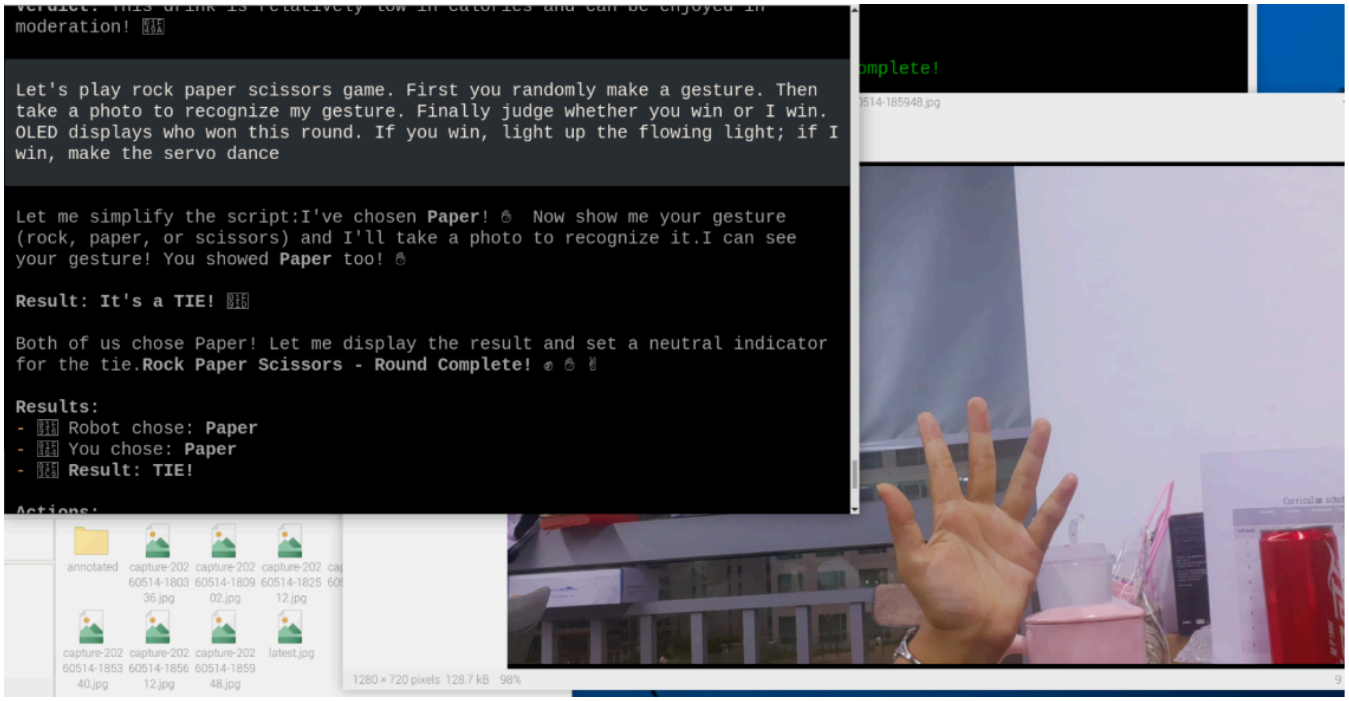
5.2 TUI Solution

If you have completed the TUI self-check in the Prerequisites section, directly enter terminal text dialogue:

```
openclaw tui
```

Then input project-related commands, for example:

- Let's play Rock-Paper-Scissors, you randomly make a gesture first, then take a photo to recognize my gesture, finally judge whether you won or I won, OLED displays who won this round, if you win light the running lights, if I win the servo performs a dance

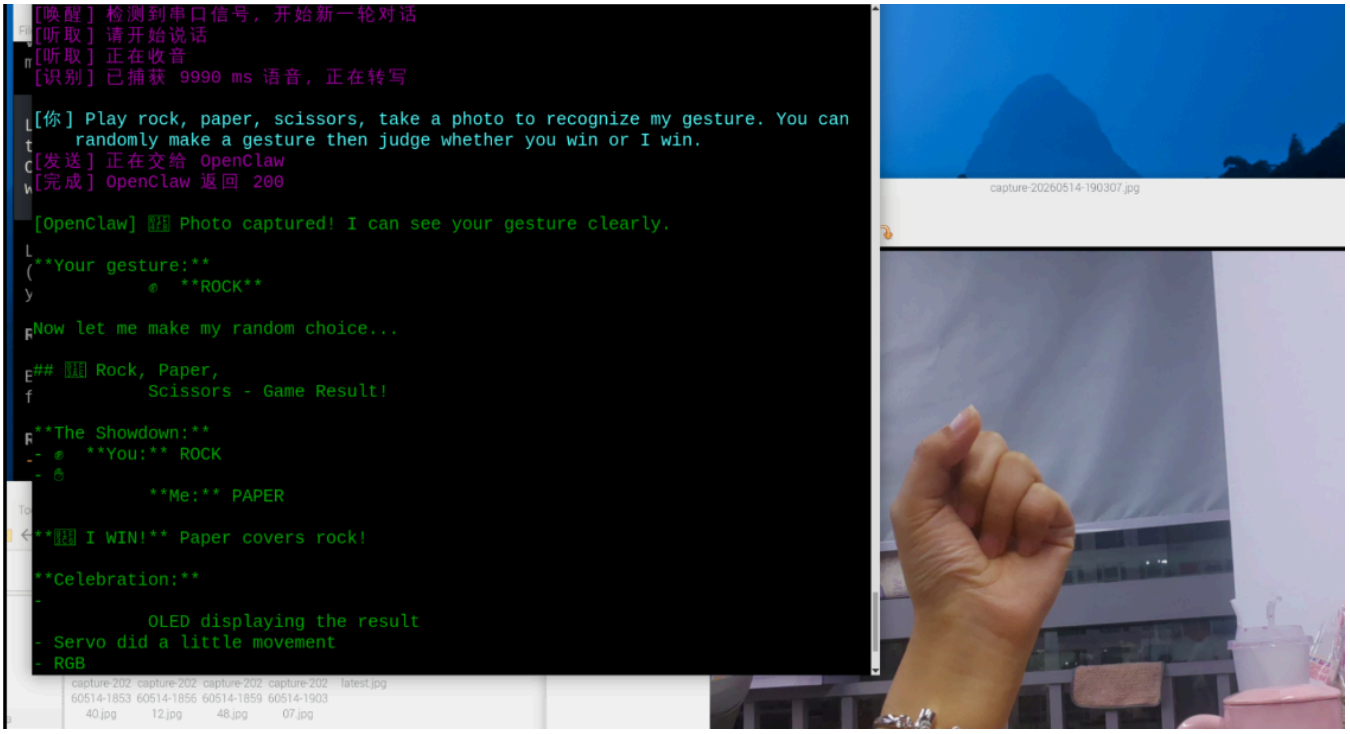


5.3 Voice Code Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- Let's play Rock-Paper-Scissors, take a photo to recognize my gesture, you can randomly make a gesture, then judge whether you won or I won, OLED displays who won this round



6. Project Implementation

Note: The following demonstrations are all conducted using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Scenario 1: Gesture Recognition

Experiment Objective

Recognize the gesture currently made by the player, such as rock, scissors, or paper.

Effect Description

OpenClaw will first take a photo, then identify the gesture type based on the image. This step provides the input for the entire game.

Example Commands

- Recognize the gesture I'm making now
- I've already made my move, check it for me
- Check if this is rock, scissors, or paper

6.2 Scenario 2: AI Move and Determination

Experiment Objective

After recognizing the player's gesture, have OpenClaw provide the AI's move result and determine the winner.

Effect Description

This step best demonstrates the "visual input + dialogue reasoning" feature. OpenClaw not only recognizes the player but also makes its own move, providing its own move and result explanation.

Example Commands

- Let's play Rock-Paper-Scissors, take a photo to recognize my gesture, you can randomly make a gesture, then judge whether you won or I won, OLED displays who won this round

6.3 Scenario 3: OLED Scoreboard

Experiment Objective

Have the OLED display the current round result or cumulative score.

Effect Description

The screen can display `Win`, `Lose`, `Draw`, or simple scores like `You 1 : 0 AI`, making the project more like a complete mini-game.

Example Commands

- Display this round's result on the OLED
- Write the score to the screen
- Display "Round 1 You Win"

6.4 Scenario 4: RGB Result Feedback

Experiment Objective

Use RGB colors to quickly express win/loss results.

Effect Description

For example, light green for a win, red for a loss, blue for a draw. This way, you can instantly see the round's result without reading any text.

Example Commands

- Light green if I win
- Light red if I lose
- If you win light the running lights, if I win the servo performs a dance